# Routed Detector Ensembles with Noise-Template Verification for Noise Event Detection in Crowd-Sourced Indic Speech

Shah Ahmed Shakir Abu Asim Khan*, Sajid Javid†, Osama Zaheer‡, Ahraz Shamim§
*Aligarh Muslim University †IIIT Delhi ‡Binary Semantics §Vecmocon Technologies
Team CodeAMU, IndoML 2026 Datathon, Track 1 Contact: sajidj@iiitd.ac.in

***Abstract*—Crowd-sourced phone recordings such as the Vaani corpus contain frequent non-speech noise events whose exact boundaries are needed to curate data and to remove the noise. Track 1 of the IndoML 2026 Datathon poses this as sound event detection with a tolerance of only 20% of the event duration on both edges. Most training clips carry unverified or clip-level labels only, and the test set mixes natural recordings with organiser-synthesised clips. We combine frame-level detectors on Whisper, WavLM and BEATs encoders, trained by a multi-tier objective with explicit onset/offset heads, a learned router that sends each clip to a synthetic or a natural ensemble with its own decoding, and a stage that localises validation-derived noise templates by cross-correlation and keeps only energy-consistent matches. The system ranked 1st on the final leaderboard with Combined 1.6173 (Event F1 0.7895, Segment Dice 0.8278).**

***Index Terms*—sound event detection, weak supervision, boundary detection, template matching, ensembles, Vaani**

## I. Introduction

*Motivation.* Large crowd-sourced corpora are central to speech technology for low-resource Indian languages. Vaani [1] records such speech on phones in more than 100 districts, so clips often contain non-speech events (traffic, animals, children, music, machines) that overlap or interrupt the speaker. Locating them with accurate boundaries enables filtering and relabelling of training data, robustness analysis of recognisers, and targeted removal of the noise, the subject of Track 2 of the same datathon [2].

*Task.* Track 1 asks for the onset and offset of every noise event in a clip. Event F1 counts a prediction only if both edges lie within $\max(0.2\,d,\ 50\,\mathrm{ms})$ of a reference event of duration $d$ (one-to-one matching); Segment Dice measures overlap on a 10 ms grid, averaged over clips; the score is their sum. The test set (5,517 clips, 10.8 h) mixes natural recordings with clips synthesised from clean speech and pasted noise.

*Challenges and research gaps.* Four properties distinguish the task from standard sound event detection (SED) [3]. (i) *Heterogeneous supervision:* only 12% of the training clips have verified timestamps, 68% unverified timestamps and 20% clip-level labels; SED systems usually assume one label quality. (ii) *Strict boundaries:* for typical 0.25–1 s events the tolerance is 50–200 ms per edge, tighter than the 200 ms collars common in SED evaluation [4], so presence-only training blurs edges. (iii) *Mixed domains:* synthesised and natural clips differ systematically in background statistics and splice artefacts, so a single decoder misplaces edges in both. (iv) *Template reuse:* synthesised clips paste recordings from a finite noise pool, a structure that learned detectors ignore but that, as in audio fingerprinting [5], permits exact localisation.

*Contributions.* (1) A multi-tier training objective that uses all three label tiers together with dedicated onset/offset heads, applied to 24 detectors with Whisper, WavLM, BEATs and CRNN front ends; (2) a learned synthetic/natural router with branch-specific ensembles and decoding; (3) a verification stage that localises validation-derived noise templates by cross-correlation and keeps only energy-consistent matches; (4) a cross-fitted validation protocol for every tuning decision. *Final standing:* **1st place**, Combined **1.6173** (Event F1 0.7895, Segment Dice 0.8278; natural clips 0.5197/0.7256, synthetic clips 0.9981/0.9966), ahead of the second-ranked entry (1.6168).

## II. Related Work

Modern SED combines convolutional and recurrent layers [6] trained from strong and weak labels, with attention pooling turning frame scores into clip scores [7]; synthetic soundscapes with exact labels are a standard source of boundary supervision [8]. Large pretrained speech and audio encoders [9], [10], [11] with learned layer weighting [12] now provide the strongest front ends. Template matching by generalised cross-correlation [13] and audio fingerprinting [5] find repeated recordings exactly, but have not been combined with learned detectors for SED. Our system uses learned detectors for unknown noise and template verification where the noise comes from a known pool.

## III. Training Data and Preprocessing

### A. Vaani Noise Event Dataset

We train on the Vaani Noise Event Dataset [14] (ARTPARK-IISc/Vaani-Noise-Event-Dataset), derived from Vaani [1]: 90,635 clips (154.6 h) with 11,111 *gold* (verified timestamps), 61,640 *silver* (unverified timestamps) and 17,884 *bronze* clips (clip-level labels only); category strings are mapped to seven classes. Audio is decoded to 16 kHz mono, crops are at most 10 s and frame targets use a 20 ms grid (10 ms for two *hr* models). Two gold hold-out sets (1,111 and 3,999 clips) serve for model selection.

### B. Official validation release

The 2,501 validation clips (1,806 natural, 695 synthetic with clean references) are the only additional data. Split into halves A/B (1,259/1,242 clips), they provide extra gold data for cross-fitted fold models (gold + one half, scored on the other) and three *full-gold* models (all 13,612 gold and validation clips), train the router and boundary regressor, tune post-processing out-of-fold, and supply the noise templates (Sec. IV-C).

### C. Boundary-exact augmentation

To obtain exact boundaries we synthesise 11,762 clips (21.4 h, 21,095 events) in the spirit of soundscape synthesis [8]: the 12,585 annotated event spans (4.0 h) and 5,884 event-free gaps (6.5 h) of 7,104 gold training clips form a noise and a clean bank; events are pasted at SNR $\sim \mathcal{U}(-22, -2)$ dB with 15 ms fades, and 35% of the inter-event gaps are drawn from 50–300 ms, because merging close events was the most frequent error.

## IV. Proposed Method

### A. Frame-level detectors with multi-tier supervision

Each detector maps a waveform to 20 ms frame logits through five heads on a shared sequence representation: presence; onset and offset (Gaussian targets, $\sigma = 40$ ms, at gold boundaries); a 7-way category; and a clip-level weak head with linear-softmax attention pooling [7]. Front ends (Table I) are Whisper [9] encoders with positional embeddings sliced to the clip length and WavLM [10], each with learned layer weighting [12] and a 2-layer BiGRU; a hybrid that concatenates frozen BEATs [11] features (frequency-pooled, upsampled $\times 8$) with a Whisper-medium stream; and log-mel CRNNs [6]. The objective lets each tier contribute what it reliably provides:

$$\mathcal{L} = \mathcal{L}_{\text{gold}} + \lambda_s \mathcal{L}_{\text{silver}} + 0.3\,\mathcal{L}_{\text{cat}} + 0.5\,\mathcal{L}_{\text{weak}} + \mathcal{L}_{\text{on}} + \mathcal{L}_{\text{off}}, \quad (1)$$

where all terms are binary cross-entropies; presence terms use gold or silver timestamps ($\lambda_s = 0.25$, 0 for gold-only fine-tuning), boundary terms use gold clips only, and the weak term is the only signal from bronze clips. Base models are trained with AdamW [15] (head and encoder learning rates $5 \cdot 10^{-4}$–$10^{-3}$ and $6 \cdot 10^{-6}$–$3 \cdot 10^{-5}$, encoder frozen for one epoch, bf16), then fine-tuned on gold data, on gold + one validation half, or on all gold + validation data.

### B. Routing and branch-specific decoding

Synthesised clips carry characteristic traces between events: a quiet, spectrally flat background, little high-band energy in quiet frames, a steady floor and energy jumps at splice points. A random forest [16] on 24 such statistics, trained on the `syntheticData` flags of the validation release, sends 2,080 of the 5,517 test clips to the synthetic branch. The *natural branch* averages 16 detectors with out-of-fold weights, blends in the full-gold models and two CRNNs, and decodes with hysteresis thresholds (0.57/0.48), event splitting at onset/offset peak pairs, duration-dependent edge snapping, a gradient-boosting [17] offset regressor, duplicate suppression and presence-based false-alarm rules. The *synthetic branch* averages 10 detectors with its own thresholds and snapping, as synthetic edges are sharp and its errors are mostly merged neighbours.

TABLE I
THE 24 DETECTORS. HUGGING FACE REVISIONS IN PARENTHESES. SYN: ALL TIERS + SYNTHETIC CLIPS; SPLIT: ALL TIERS MINUS A GOLD HOLD-OUT; FT: GOLD-ONLY FINE-TUNE; A/B: GOLD + ONE VALIDATION HALF; FULL: ALL GOLD + VALIDATION.

| Encoder (revision) | Params | # | Training variants |
|---|---|---|---|
| Whisper large-v3 (06f233f) | 640.6M | 7 | split, split+FT, syn, syn+FT, A, B, full |
| Whisper large-v2 (ae46427) | 640.4M | 2 | split+FT, full |
| Whisper medium (abdf7c3) | 310.4M | 5 | syn, A, B, syn and syn+FT at 10 ms |
| Whisper small (973afd2) | 90.9M | 3 | syn, A, B |
| WavLM Base+ (4c66d48) | 97.2M | 1 | split+FT |
| BEATs (172e407) + Whisper-m. | 405.1M | 2 | split+FT, full |
| Log-mel CRNN | 0.8/4.4M | 4 | split+FT, syn ($\times 2$), syn+FT |

### C. Noise-template verification (disclosure)

The validation release exposes the noise pool: *mixture* − *clean* at the labelled timestamps yields 2,213 events, 1,181 distinct recordings (4,118 templates with the labelled events of natural validation clips). After pre-emphasis (0.97) and resampling to 8 kHz, each test clip is cross-correlated with every unit-norm template by FFT, normalised by the local signal energy [13]; up to three peaks per template with correlation $\geq 0.20$ are kept and matches are chosen greedily by explained energy. A match at lag $t$ implies the span $[t, t+\ell]$, which equals the annotation for 1,925 of 1,926 cross-clip copies in the validation set at correlation $\geq 0.6$ (median edge error 0.3 ms). Truncated or spurious matches are rejected by an energy test: after subtracting the template with a fitted gain, some 10 ms frame must lose at least 1 dB and the first and last 100 ms must not gain energy (thresholds per correlation band; at 0.30–0.45, at least 30% of the template frames must be explained). Accepted spans snap, replace or add events in synthetic clips and, since the pool was cut from natural recordings, in natural clips (correlation $\geq 0.7$, 94–96% exact on validation). The organisers confirmed in writing that validation-derived noise waveforms may be located in test audio by cross-correlation to determine Track 1 event boundaries, provided the method is disclosed, as done here.

### D. Refinement

Seventy-five synthetic predictions overlap no template. Our Track 2 enhancer [2] serves as a noise meter: the share of energy it removes over an event, $D = 10 \log_{10}(E_{\text{removed}}/E_{\text{mixture}})$, has a median of $-2$ dB on true noise and $-64$ dB on noise-free speech, so the 28 events with $D < -12$ dB are dropped. Because Segment Dice scores an empty clip as 0, each of the 31 natural clips without predictions receives one event over the ensemble's activity envelope (first to last frame with $p \geq 0.20$).

TABLE II
LEADERBOARD RESULTS ON THE 5,517 TEST CLIPS AS THE SYSTEM WAS BUILT UP.

| System | Comb. | F1 | Dice | $F1_{nat}$ | $F1_{syn}$ |
|---|---|---|---|---|---|
| Routed ensembles | 1.5985 | .7773 | .8212 | .5096 | .9865 |
| + synthetic duration snapping | 1.6001 | .7788 | .8213 | .5096 | .9892 |
| + template snaps, synthetic | 1.6066 | .7812 | .8254 | .5096 | .9936 |
| + natural templates, post-processing | 1.6134 | .7879 | .8255 | .5193 | .9946 |
| + energy test, noise meter | 1.6158 | .7897 | .8262 | .5193 | .9980 |
| + verified spans in natural clips | 1.6167 | .7903 | .8264 | .5208 | .9981 |
| **+ empty-clip handling (final, 1st)** | **1.6173** | .7895 | .8278 | .5197 | .9981 |

### E. Implementation

PyTorch 2.11 [18], `transformers` [19] and scikit-learn [20] on NVIDIA H200 GPUs (141 GB; 71 GB MIG slices for fine-tuning); the 24 detectors process 10.9 h of audio in about 50 min. Code, configurations and all checkpoints: https://github.com/sajidjavid222/CodeAMU_Track1, https://huggingface.co/sajidjavid/CodeAMU-IndoML2026-Track1.

## V. RESULTS AND CONCLUSION

### A. Protocol and main results

All decisions were taken on cross-fitted validation (fold models scored only on the unseen half; a change kept only if both halves improved), with the leaderboard as the final check. Table II decomposes the final system. Routing is the decisive design choice: before it, our submissions scored 1.15–1.17 with synthetic-subset F1 below 0.38, while routed ensembles reach 1.60. Template verification then nearly solves the synthetic part, raising its F1 from 0.989 to 0.998 and its Dice from 0.983 to 0.997; templates that recur in natural recordings add a smaller gain (+0.0026 Combined from the natural merge, +0.0008 from energy-verified spans). Individual detectors score 1.38–1.45 Combined on held-out validation, so most of the advantage comes from the system built around them.

### B. Analysis

Natural clips remain the bottleneck (F1 0.52, against 0.998 on synthetic clips). Out of fold, 43% of reference events are matched, 46% are detected but miss the boundary tolerance (offsets more often than onsets) and only 10.5% are missed entirely. Edge errors of environmental sounds (vehicle, animal, baby) are scattered rather than biased (median 90–150 ms against 23–39 ms for human non-speech), which points to inherently ambiguous edges of gradual sounds rather than a correctable model bias: an oracle that fixed both edges of these near-misses would add +0.52 Combined, removing all pure false alarms only +0.02. Because Dice is averaged per clip, deleting an overlapping prediction usually costs more Dice than it gains F1, whereas giving each empty clip one event helps.

### C. Limitations and future work

Template verification requires the evaluation noise to come from a pool the validation release exposes; on unseen noise only the learned branches apply (synthetic F1 0.989 before verification). For natural recordings, boundary models that express uncertainty or learn annotator conventions are the most promising direction.

### D. Conclusion

Multi-tier supervised detectors, domain routing and exact template verification together gave the winning Track 1 system (1st place, Combined 1.6173). From cached detector outputs the released pipeline regenerates the final submission in 5,516 of 5,517 clips (one edge differs by 0.5 ms), and re-running four of the detectors reproduces their outputs bit for bit.

## REFERENCES


[1] S. Pulikodan, A. Singh, A. Basu, N. Desai, P. Kumar J, P. D. Bhat *et al.*, "VAANI: Capturing the language landscape for an inclusive digital India," arXiv:2603.28714, 2026.

[2] S. A. S. A. A. Khan, S. Javid, O. Zaheer, and A. Shamim, "Known-noise subtraction and event-conditioned complex masking for noise event removal in crowd-sourced Indic speech," CodeAMU technical report, IndoML 2026 Datathon, Track 2, 2026, https://github.com/sajidjavid222/CodeAMU_Track2.

[3] A. Mesaros, T. Heittola, T. Virtanen, and M. D. Plumbley, "Sound event detection: A tutorial," *IEEE Signal Process. Mag.*, vol. 38, no. 5, pp. 67–83, 2021.

[4] A. Mesaros, T. Heittola, and T. Virtanen, "Metrics for polyphonic sound event detection," *Applied Sciences*, vol. 6, no. 6, p. 162, 2016.

[5] A. Wang, "An industrial-strength audio search algorithm," in *Proc. ISMIR*, 2003.

[6] E. Çakır, G. Parascandolo, T. Heittola, H. Huttunen, and T. Virtanen, "Convolutional recurrent neural networks for polyphonic sound event detection," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 25, no. 6, pp. 1291–1303, 2017.

[7] Y. Wang, J. Li, and F. Metze, "A comparison of five multiple instance learning pooling functions for sound event detection with weak labeling," in *Proc. ICASSP*, 2019.

[8] N. Turpault, R. Serizel, A. Parag Shah, and J. Salamon, "Sound event detection in domestic environments with weakly labeled data and soundscape synthesis," in *Proc. DCASE Workshop*, 2019.

[9] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proc. ICML*, 2023.

[10] S. Chen *et al.*, "WavLM: Large-scale self-supervised pre-training for full stack speech processing," *IEEE J. Sel. Topics Signal Process.*, vol. 16, no. 6, pp. 1505–1518, 2022.

[11] S. Chen, Y. Wu, C. Wang, S. Liu, D. Tompkins, Z. Chen, and F. Wei, "BEATs: Audio pre-training with acoustic tokenizers," in *Proc. ICML*, 2023; checkpoint https://huggingface.co/WeiChihChen/BEATs_iter3_plus_AS2M_finetuned_on_AS2M_cpt2.

[12] S.-w. Yang *et al.*, "SUPERB: Speech processing universal performance benchmark," in *Proc. Interspeech*, 2021.

[13] C. Knapp and G. Carter, "The generalized correlation method for estimation of time delay," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 24, no. 4, pp. 320–327, 1976.

[14] ARTPARK-IISc, "Vaani Noise Event Timestamps," Hugging Face dataset `ARTPARK-IISc/Vaani-Noise-Event-Dataset`, 2026.

[15] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. ICLR*, 2019.

[16] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[17] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Ann. Statist.*, vol. 29, no. 5, pp. 1189–1232, 2001.

[18] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Proc. NeurIPS*, 2019.

[19] T. Wolf *et al.*, "Transformers: State-of-the-art natural language processing," in *Proc. EMNLP: System Demonstrations*, 2020.

[20] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.